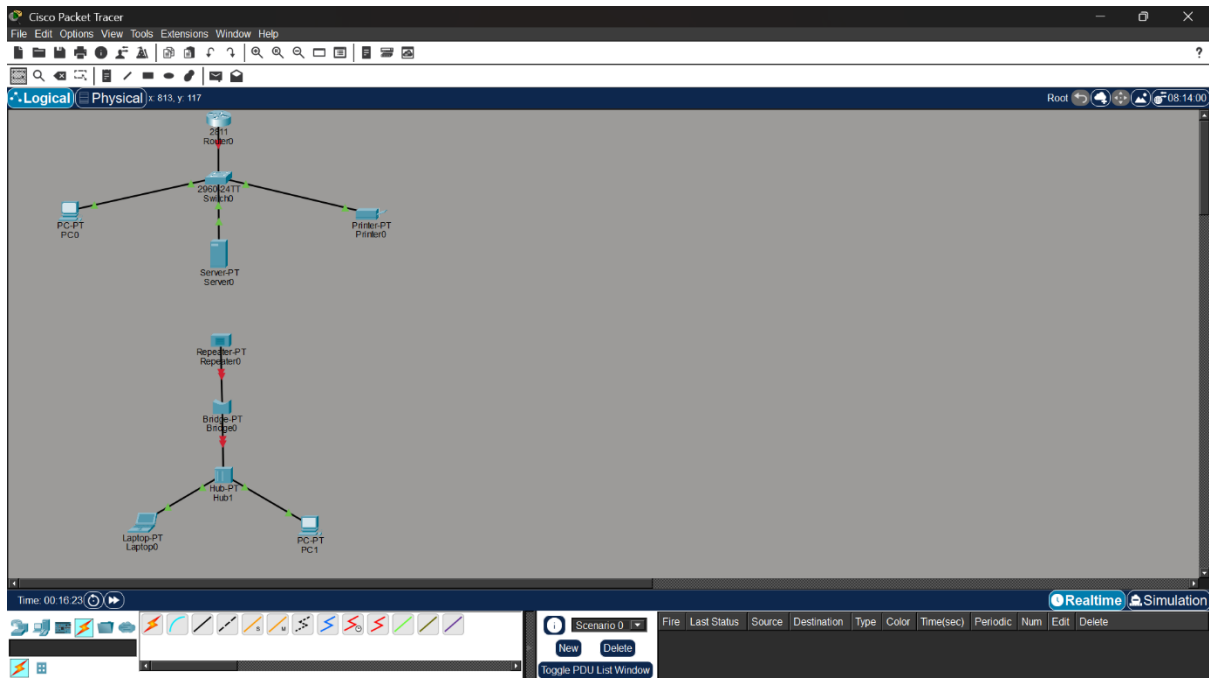
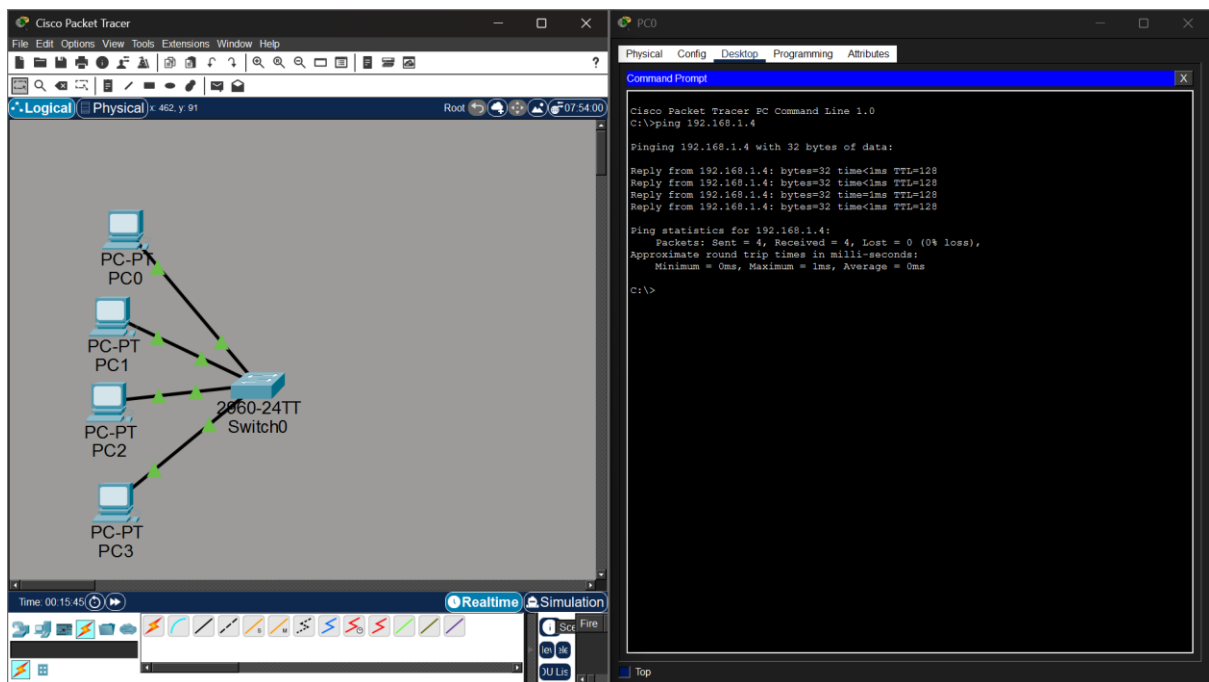


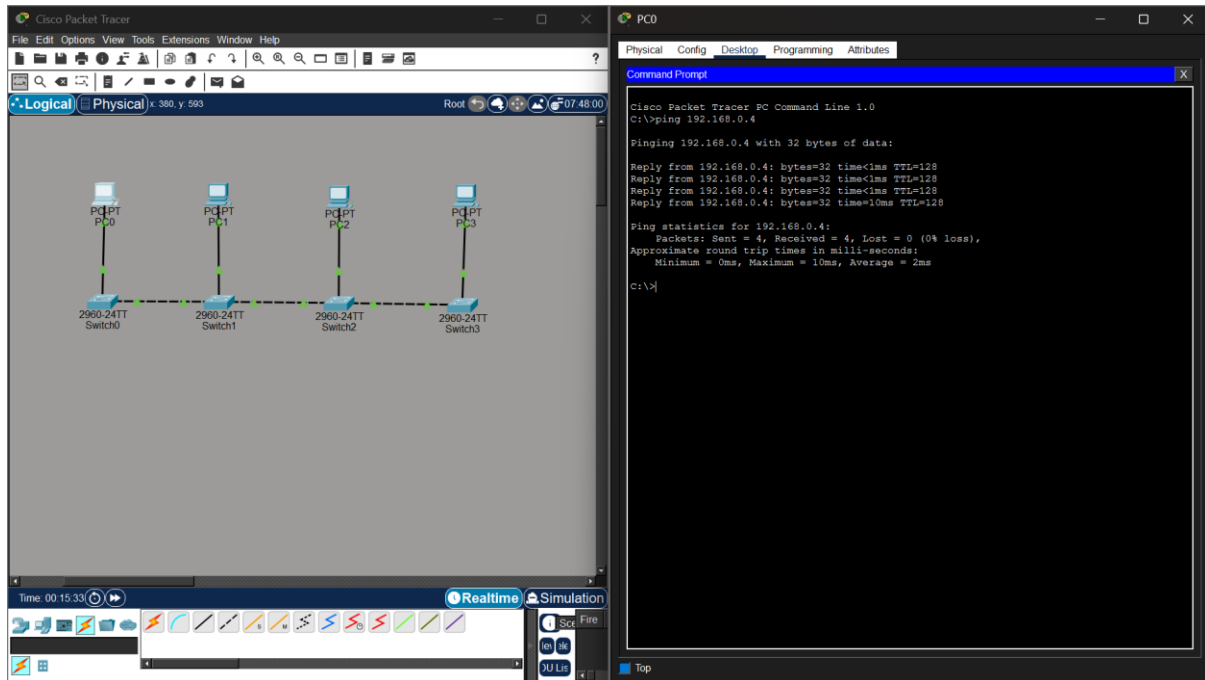
Experiment-01-Network-Components



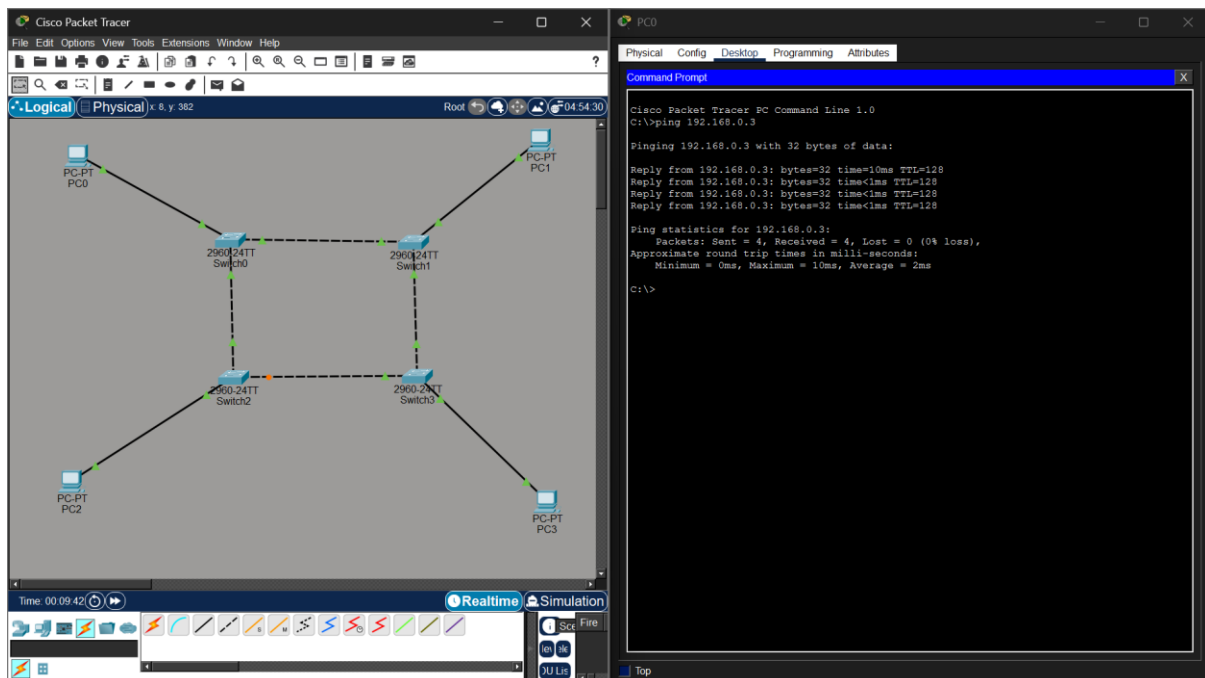
Experiment-02-Star-Topology



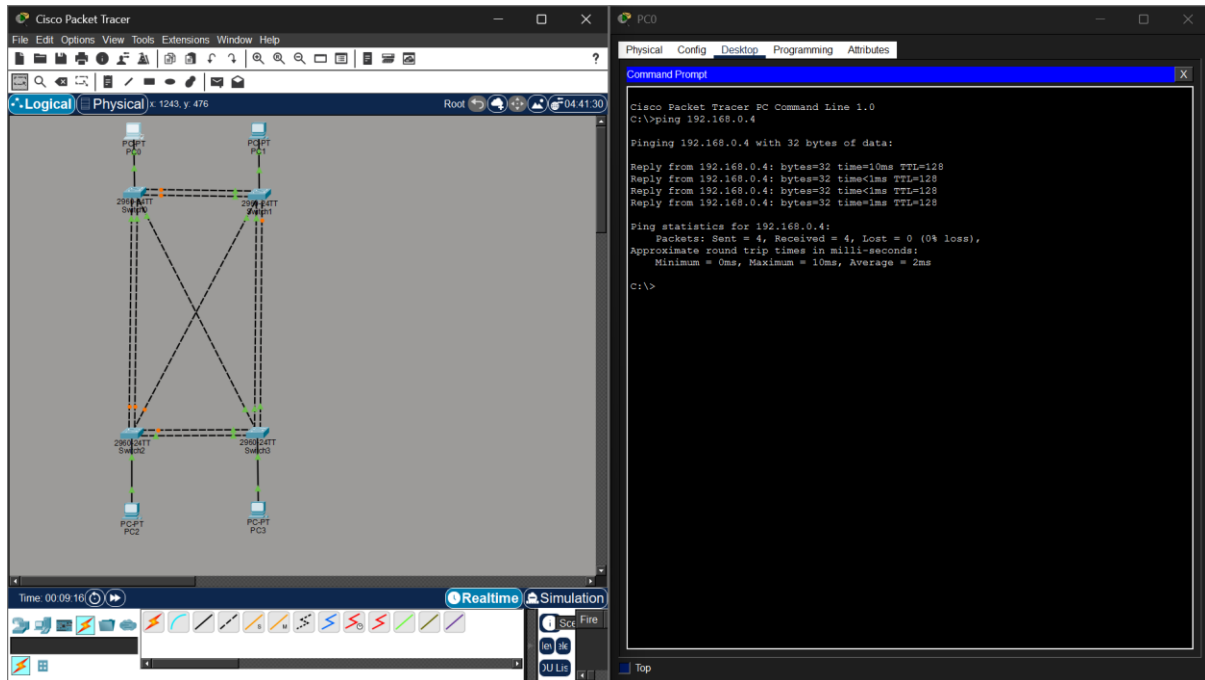
Experiment-03-Bus-Topology



Experiment-04-Ring-Topology



Experiment-05-Mesh-Topology



Experiment-06-Tree-Topology

Cisco Packet Tracer interface showing a tree topology. The central device is a 2503 Switch, which is connected to two 2950 Switches. The left 2950 Switch is connected to three PCs (PC0, PC1, PC2), and the right 2950 Switch is connected to three PCs (PC3, PC4, PC5). The PDU List Window shows four successful ICMP events:

File	Last Status	Source	Destination	Type	Color	Time/Sec	Periodic	Num	Edit	Delete
	Successful	PC0	PC3	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC1	PC4	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC2	PC5	ICMP		0.000	N	2	(edit)	(delete)
	Successful	PC6	PC0	ICMP		0.000	N	3	(edit)	(delete)

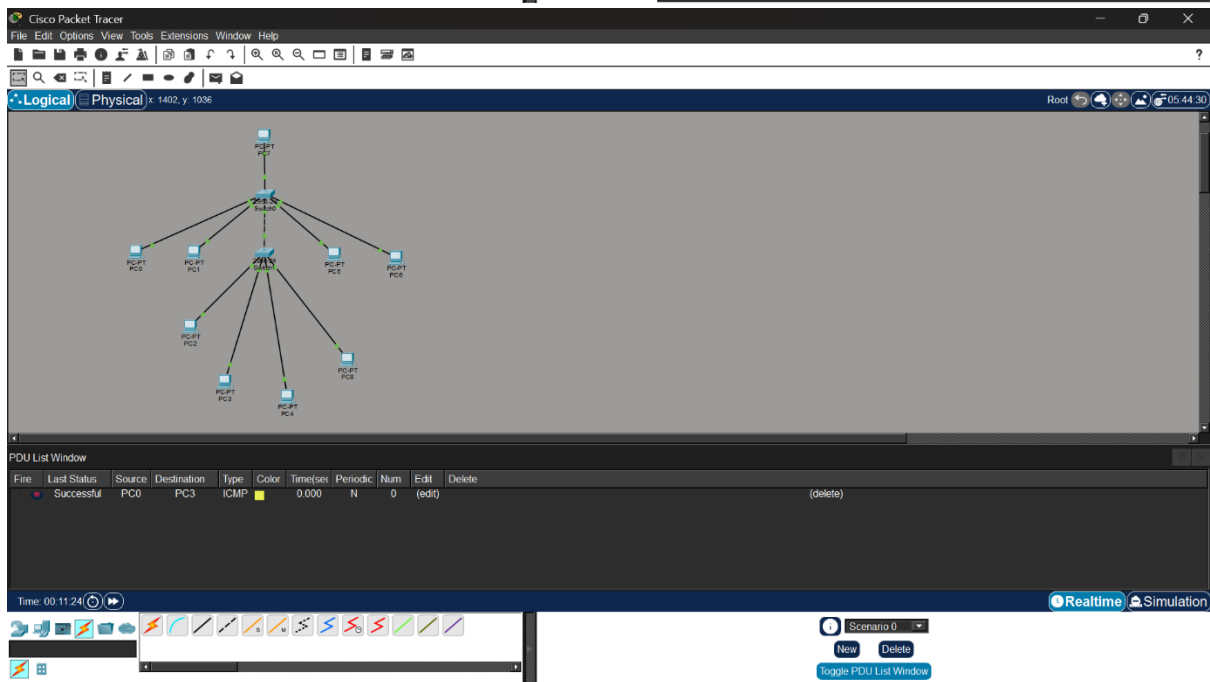
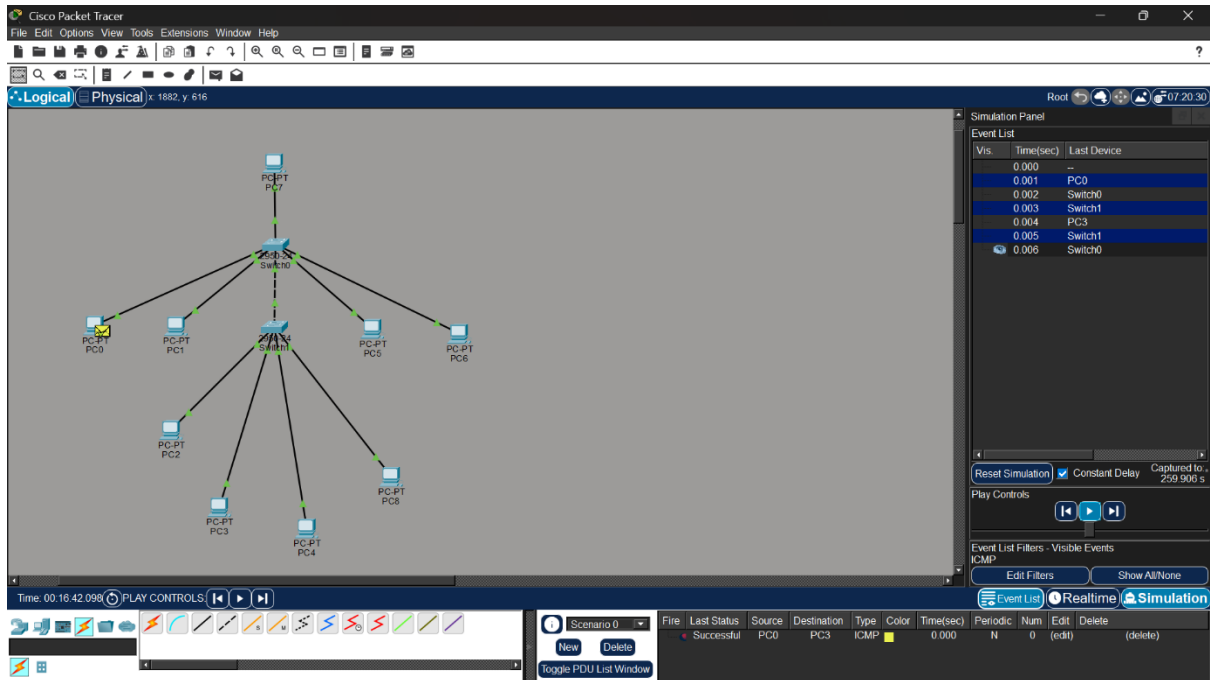
Time: 00:17:34.731

Cisco Packet Tracer interface showing the same tree topology. The PDU List Window shows four successful ICMP events:

File	Last Status	Source	Destination	Type	Color	Time/Sec	Periodic	Num	Edit	Delete
	Successful	PC0	PC3	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC1	PC4	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC2	PC5	ICMP		0.000	N	2	(edit)	(delete)
	Successful	PC6	PC0	ICMP		0.000	N	3	(edit)	(delete)

Time: 00:14:57

Experiment-07-Hybrid-Topology



Experiment-08-ARP-Analysis

The image displays two screenshots of the Cisco Packet Tracer interface, illustrating a network setup and the process of analyzing ARP requests.

Top Screenshot: The network topology shows a central Hub0 connected to six PCs (PC0, PC1, PC2, PC3, PC4, PC5). The PDU List Window is open, showing a successful ICMP packet from PC0 to PC3. The time is 00:12:00.

Bottom Screenshot: The network topology is the same, but the PDU List Window shows a successful ICMP packet from PC0 to PC3. The time is 01:08:51.264. The Simulation Panel is open, showing the Event List and Play Controls. The Event List shows a series of events, including ARP requests and replies, occurring at various times. The Play Controls show the simulation is running in Realtime mode.

Network Topology:

```
graph TD
    Hub0[Hub0] --- PC0[PC0]
    Hub0 --- PC1[PC1]
    Hub0 --- PC2[PC2]
    Hub0 --- PC3[PC3]
    Hub0 --- PC4[PC4]
    Hub0 --- PC5[PC5]
```

PDU List Window (Top Screenshot):

File	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC3	ICMP		0.000	N	0	(edit)	(delete)

Simulation Panel (Bottom Screenshot):

Event List:

Vis.	Time(sec)	Last Device
	0.000	
	0.001	PC0
	0.002	Hub0
	0.002	Hub0
	0.002	Hub0
	0.002	Hub0
	0.002	Hub0
	0.003	PC3
	0.004	Hub0
	0.004	Hub0
	0.004	Hub0
	0.004	Hub0

Play Controls: Reset Simulation, Constant Delay, Captured to: 135.320 s

Event List Filters - Visible Events: ACL Filter, ARP, Bluetooth, CAPWAP, CDP, DHCPv6, DTP, EAPOL, EIGRPv6, FTP, H.323, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IGMP, IPsec, ISAKMP, IoT, IoT, TCP, LACP, LLDP, MODBUS, Meraki, NDP, NETFLOW, NTP, OSPFv6, PAgP, POP3, PPP, PPPoE, PTP, Protocol, RADIUS, REP, RFPing, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP

Edit Filters, Show All/None

Event List, Realtime, Simulation

Experiment-09-CSMA-CD-CSMA-CA

The image displays two screenshots of the Cisco Packet Tracer interface, illustrating network configurations and command-line outputs for CSMA-CD and CSMA-CA experiments.

Top Screenshot: The main window shows a network topology with a central switch (Switch0) connected to three PCs (PC0, PC1, PC2). The Simulation Panel on the right shows the Event List with the following entries:

Vis.	Time(sec)	Last Device
0.000	-	-
0.001	PC0	
0.002	Switch0	
0.003	PC1	
0.004	Switch0	

The Command Prompt window shows the following output:

```
FastEthernet0 Connection:(default port)
Connection-specific DNS Suffix...:
Link-local IPv6 Address...: FE80::2D0:FFFF:FED5:9268
IPv6 Address...: ::
IPv4 Address...: 192.168.1.1
Subnet Mask...: 255.255.255.0
Default Gateway...: 0.0.0.0

Bluetooth Connection:
Connection-specific DNS Suffix...:
Link-local IPv6 Address...: ::
IPv6 Address...: ::
IPv4 Address...: 0.0.0.0
Subnet Mask...: 0.0.0.0
Default Gateway...: 0.0.0.0

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Bottom Screenshot: The main window shows the same network topology. The Simulation Panel on the right shows the Event List with the following entries:

Vis.	Time(sec)	Last Device
0.000	-	-
0.001	PC0	
0.002	Switch0	
0.003	PC1	
0.004	Switch0	

The Command Prompt window shows the following output:

```
FastEthernet0 Connection:(default port)
Connection-specific DNS Suffix...:
Link-local IPv6 Address...: FE80::2D0:FFFF:FED5:9268
IPv6 Address...: ::
IPv4 Address...: 192.168.1.1
Subnet Mask...: 255.255.255.0
Default Gateway...: 0.0.0.0

Bluetooth Connection:
Connection-specific DNS Suffix...:
Link-local IPv6 Address...: ::
IPv6 Address...: ::
IPv4 Address...: 0.0.0.0
Subnet Mask...: 0.0.0.0
Default Gateway...: 0.0.0.0

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Experiment-10-Computer-Lab

The image displays two screenshots of the Cisco Packet Tracer application. The top screenshot shows a network topology in the Logical view. A central 2960-24 Switch0 is connected to four PC-PT devices (PC0, PC1, PC2, PC3) and a 1841 Router0. The bottom screenshot shows the same network topology, but with the Command Prompt window open for PC0. The Command Prompt displays the output of the 'ipconfig' command, showing the IPv4 address 192.168.1.1, subnet mask 255.255.255.0, and default gateway 0.0.0.0. It also shows the output of the 'ping 192.168.1.254' command, indicating successful connectivity with 0% loss.

Network Topology:

- Router0 (1841) is connected to Switch0 (2960-24).
- Switch0 (2960-24) is connected to PC-PT PC0, PC-PT PC1, PC-PT PC2, and PC-PT PC3.

Command Prompt Output (PC0):

```
C:\>ipconfig

FastEthernet0 Connection (default port)

    Connection-specific DNS Suffix...: 
    Link-local IPv6 Address . . . . .: FE80::230:F2FF:FETA:2C39
    IPv6 Address . . . . .: 
    IPv4 Address. . . . .: 192.168.1.1
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: 0.0.0.0

Bluetooth Connection:

    Connection-specific DNS Suffix...: 
    Link-local IPv6 Address . . . . .: 
    IPv6 Address . . . . .: 
    IPv4 Address. . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: 0.0.0.0

C:\>ping 192.168.1.254

Pinging 192.168.1.254 with 32 bytes of data:

Reply from 192.168.1.254: bytes=32 time=12ms TTL=255
Reply from 192.168.1.254: bytes=32 time<1ms TTL=255
Reply from 192.168.1.254: bytes=32 time<1ms TTL=255
Reply from 192.168.1.254: bytes=32 time<1ms TTL=255

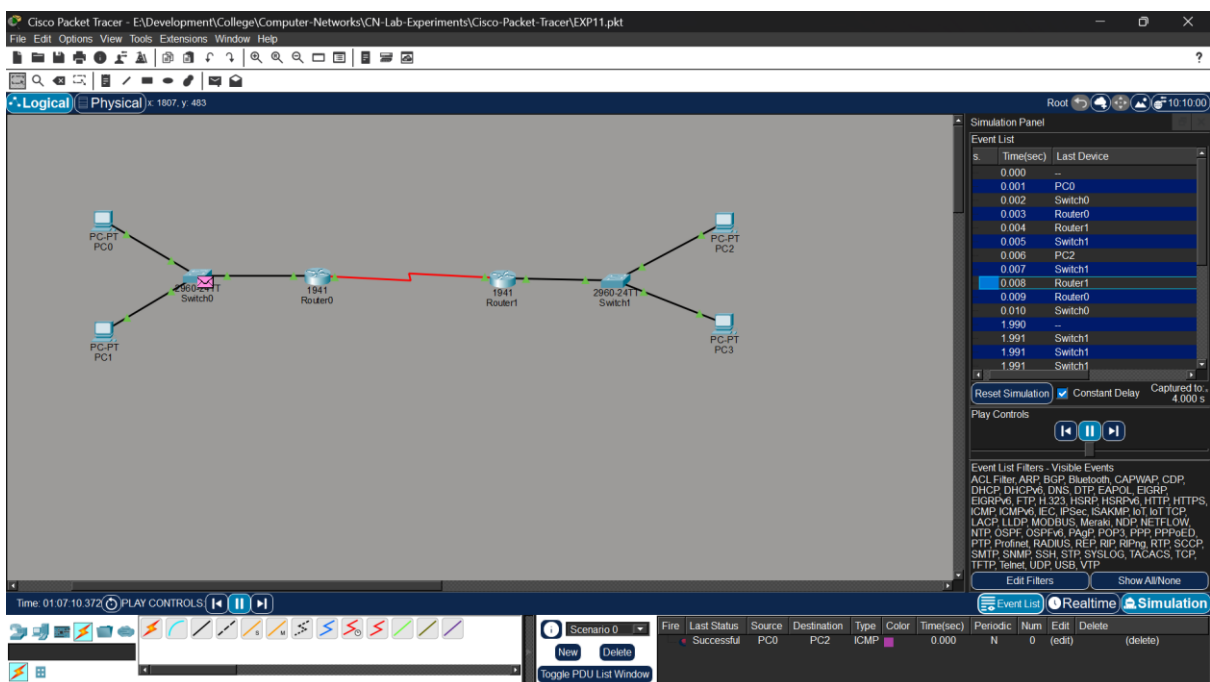
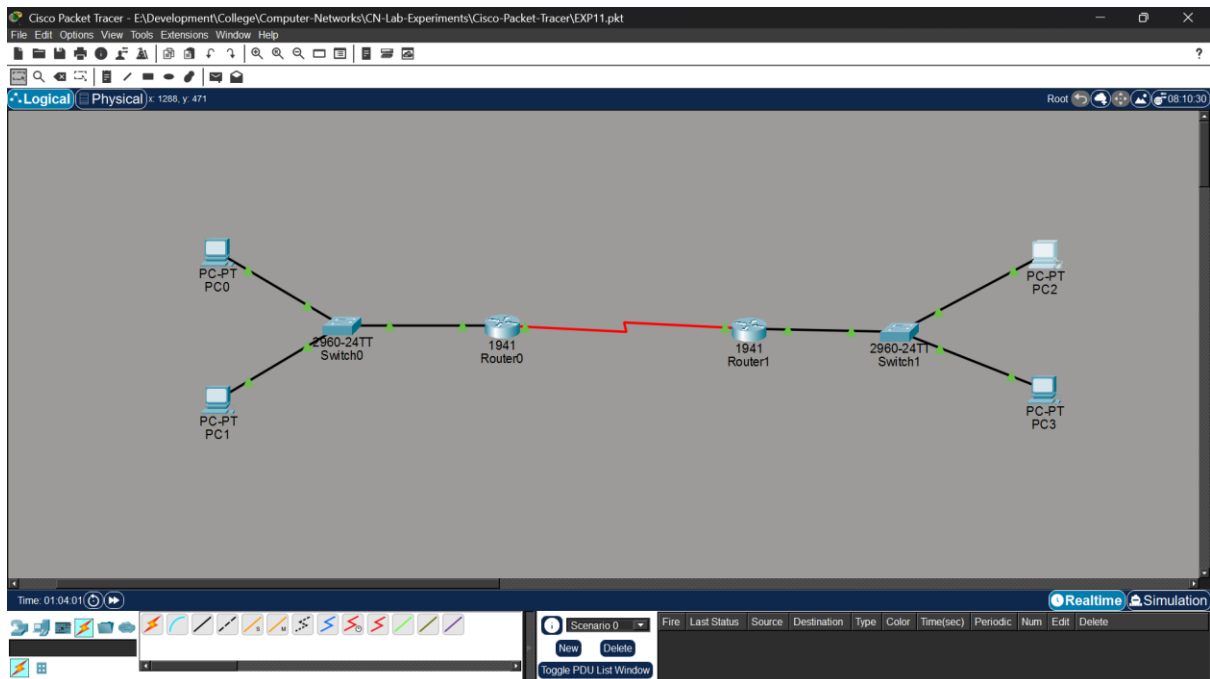
Ping statistics for 192.168.1.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 3ms

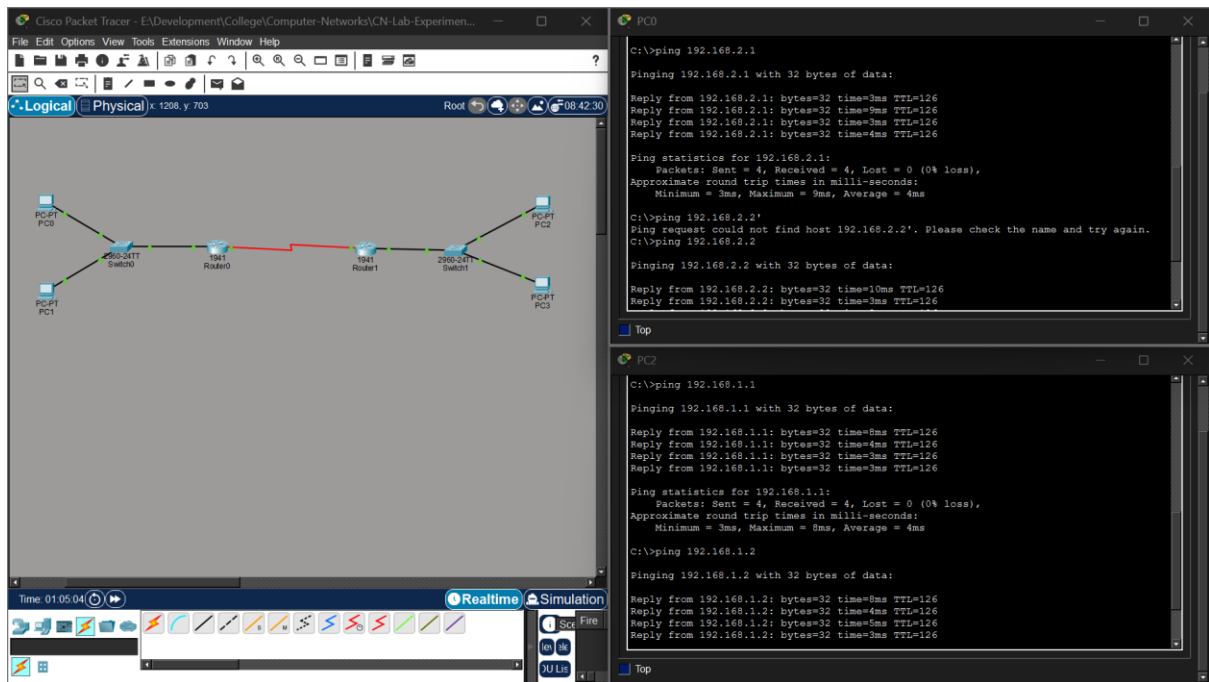
C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

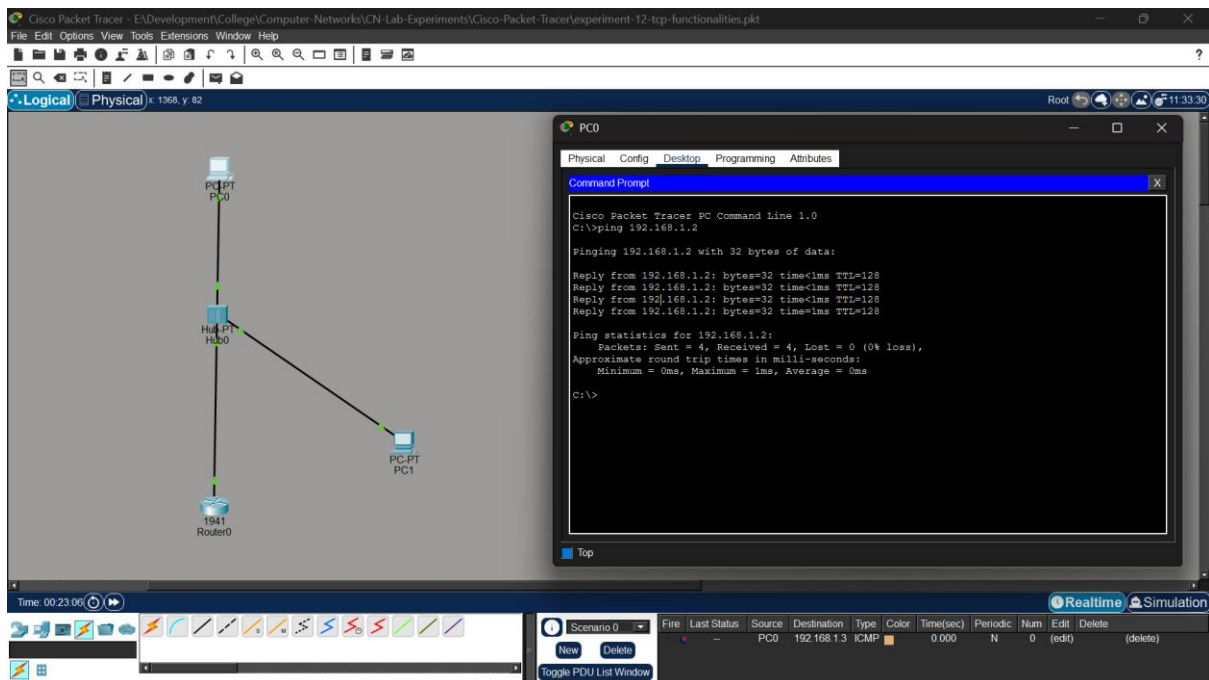
Reply from 192.168.1.4: bytes=32 time=1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
```


Experiment-11-Computer-Lab Router to Router Communication Using Static Routing





Experiment-12-Tcp-Functionalities



Experiment-13- Experiment-class-c-subnetting

The screenshot displays the Cisco Packet Tracer interface for a network experiment titled "Subnetting - Class C Addressing". The network topology consists of a central Router1 (2911) connected to two switches (2950). The left switch is connected to three PCs (PC0, PC1, PC2) and the right switch is connected to three PCs (PC3, PC4, PC5). A PC window is open, showing the Command Prompt with the following output:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.129

Pinging 192.168.10.129 with 32 bytes of data:

Reply from 192.168.10.129: bytes=32 time<1ms TTL=127
Reply from 192.168.10.129: bytes=32 time<1ms TTL=127
Reply from 192.168.10.129: bytes=32 time<1ms TTL=127
Reply from 192.168.10.129: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.129:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

The bottom status bar shows the simulation is running in Realtime mode, with a time of 00:09:18. The interface also includes a toolbar with various icons for network configuration and a bottom panel with buttons for "New", "Delete", and "Toggle PDU List Window".